

Module 04: Regularisation



Courtesy to Andrew Ng Machine Learning and Deep Learning Course DeepLearning.AI

Learning Outcomes/Agenda

- Recap – Week 3 Review
- Why Models Fail? (Generalisation)
- Underfitting vs Overfitting
- Regularization Intuition
- L1 vs L2 Regularization
- Interactive Python Demonstration
- GenAI Critique Activity
- Reflection & Key Takeaways

Module-03 Recap

- Explain major application areas of Deep Learning.
- Compare NLP, Speech Recognition, and Computer Vision applications.
- Identify datasets, models, and challenges associated with different applications.
- Critically evaluate the opportunities and risks of Deep Learning.
- Develop an application idea relevant to Assessment 1.

Recap Quiz

Question 1: Which statement best describes Natural Language Processing (NLP)?

- A. Teaching robots to walk
- B. Enabling computers to understand and process human language
- C. Detecting objects in images
- D. Predicting stock prices only

Answer: B

Discussion: Can you name an NLP application you have used in the last 24 hours?

Question 2: What is the primary input for a Computer Vision system?

- A. Audio signals
- B. Text documents
- C. Images or videos
- D. Temperature readings

Answer: C

Discussion: When humans look at an image, we immediately recognise objects. What must a computer first learn before it can recognise an object?

Recap

Question 3: Which Deep Learning application is primarily designed to convert spoken language into text?

- A. Computer Vision
- B. Natural Language Processing
- C. Speech Recognition
- D. Reinforcement Learning

Answer: C

Discussion: Why is speech recognition more difficult than simply recognising text?

Question 4: Which statement is TRUE about Deep Learning applications?

- A. The same model is always suitable for every application.
- B. Every Deep Learning problem requires identical datasets.
- C. Different applications often require specialised models and datasets.
- D. Computer Vision only works for autonomous vehicles.

Answer: C

Discussion: Why might an image recognition model not perform well on speech recognition?

Recap

Question 5: Why is human judgement still important when using Deep Learning systems?

- A. AI is always correct.
- B. AI can make decisions without data.
- C. AI outputs should be evaluated for fairness, accuracy, ethics, and real-world impact.
- D. Human judgement is no longer needed.

Answer: C

Discussion: Can you think of a situation where you would not completely trust an AI system?

• Reflection Activity (Think–Pair–Share)

Imagine you are developing an AI system for a hospital. Which application area would you choose?

- NLP
- Speech Recognition
- Computer Vision

Discuss:

- What would be the input?
- What would be the output?
- What challenges would you face?
- Would accuracy alone be enough to trust the system?

Week 3 & 4 Connection

- **Week 3:** What amazing things can Deep Learning do?
- **Week 4:** How do we ensure Deep Learning models perform reliably in the real world?



Assessment Progression

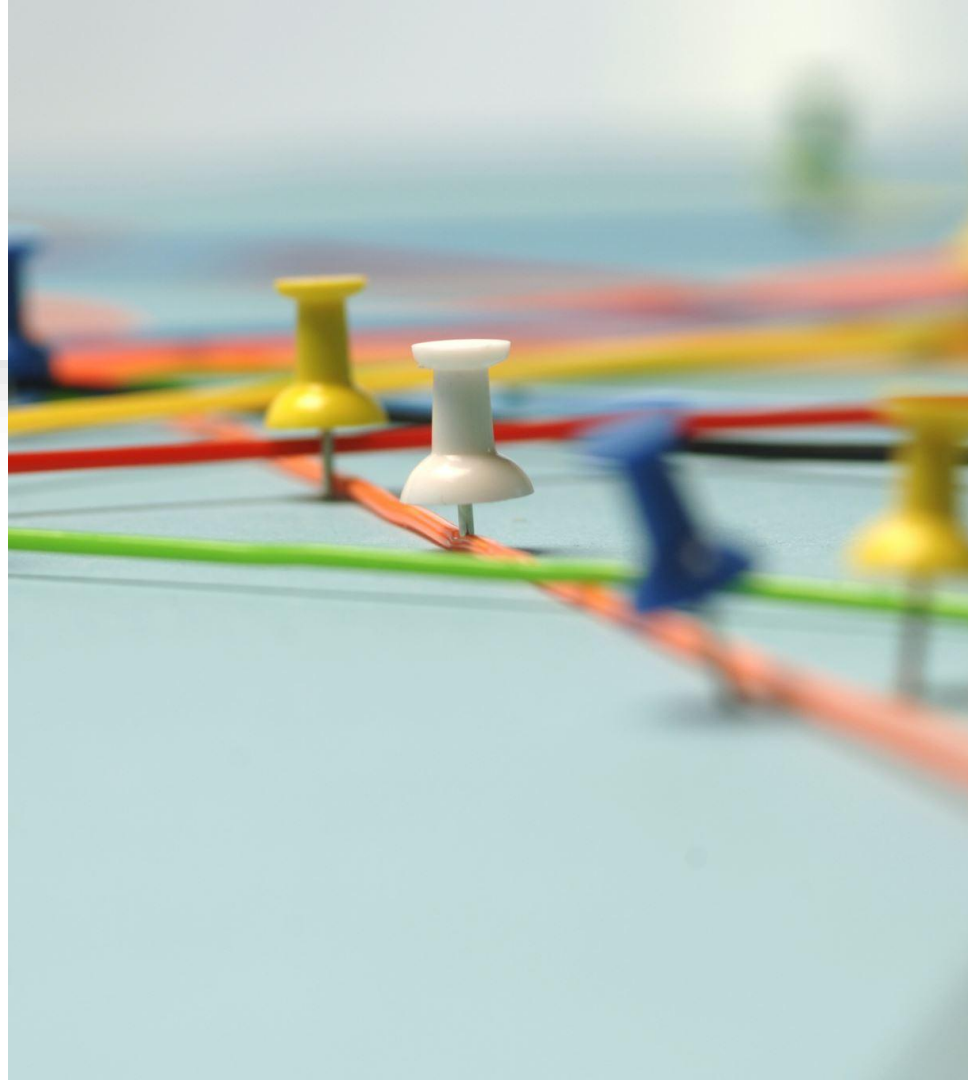
- **Assessment 1 is due at the end of Module 4.** Please refer to the assessment briefs in the assessment area for more detail.
- You are expected to spend approximately one hour reviewing Assessment 1 in this Module. You should:
 - Review and familiarize yourself with Assessment 1;
 - Enhance your understanding of the key concepts of deep learning; and
 - Review algorithms, such as N-Gram, in preparation for Assessment 1.
- ***You can prepare for this Assessment by using the learning resources of this Module,*** which include readings, videos and learning activities. These resources will enhance your understanding of the key concepts in this Module.

Why reliability in DL models Matter?

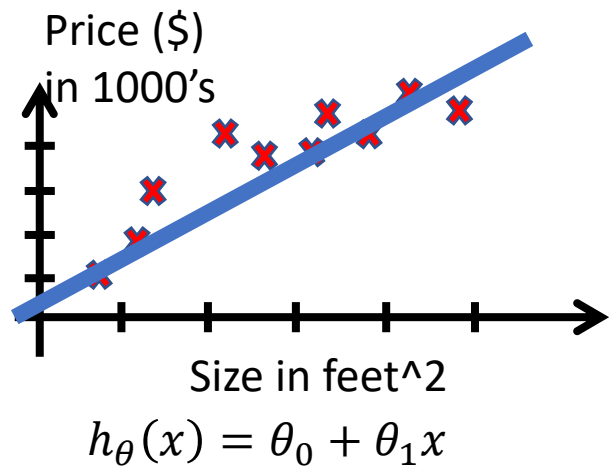
Here are three predictions:

- Training Accuracy = 100%
- Testing Accuracy = 48%
- Validation Accuracy = 51%

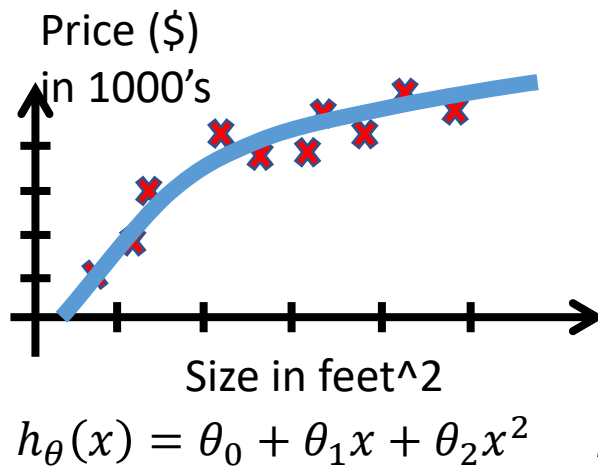
Would you deploy this model?



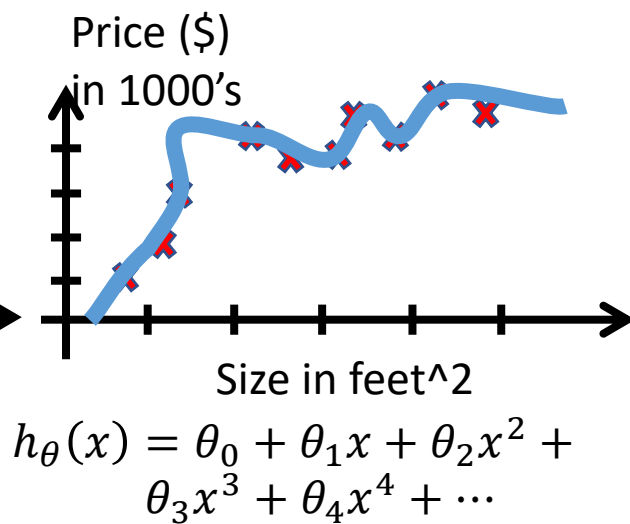
Generalisation, Underfitting, Good fit, Overfitting



Underfitting

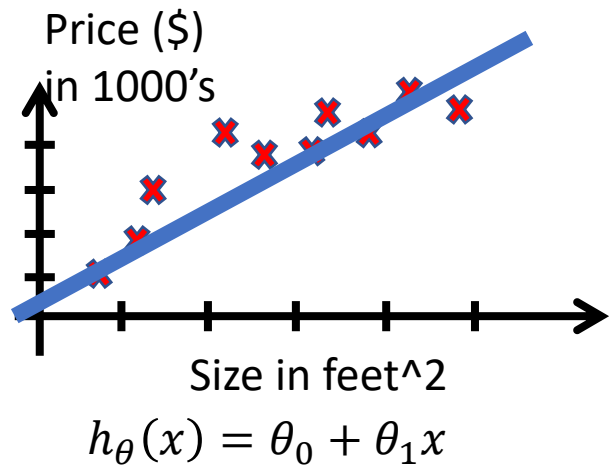


Just right



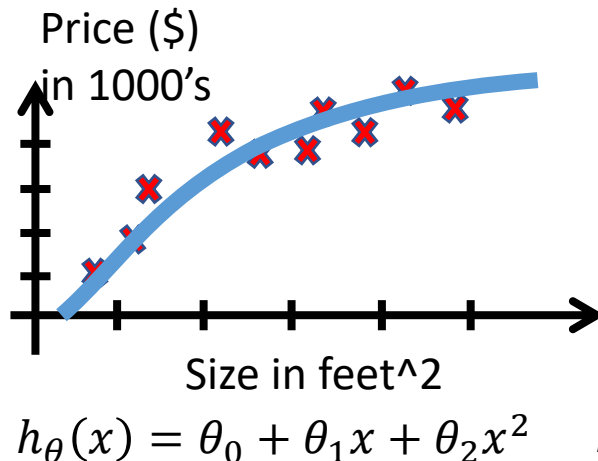
Overfitting

Generalisation, Underfitting, Good fit, Overfitting

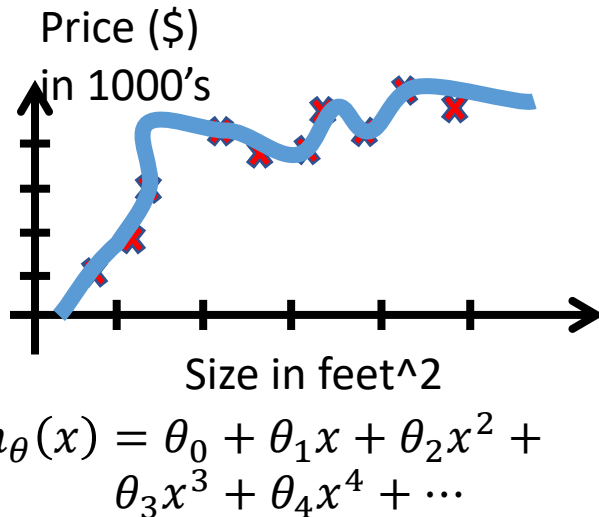


Underfitting

High bias



Just right



Overfitting

High variance

Bias-Variance Tradeoff

- **Bias:** difference between **what you expect to learn** and **truth**
 - Measures how well you expect to represent true solution
 - Decreases with more complex model
- **Variance:** difference between **what you expect to learn** and **what you learn from a particular dataset**
 - Measures how sensitive learner is to specific dataset
 - Increases with more complex model

Bias-Variance

- Suppose – you memorise every previous exam paper. Will you necessarily perform well next trimester?
 - Yes/NO?
 - Alternatively, Suppose you only study one chapter. How about your performance in the next trimester exam?
 - Yes/NO
- High Bias

Human Analogy: Exam Preparation

Student A

- Memorises every past exam question.

Student B

- Understands the concepts.

→ Who performs better when the exam changes?

→ Regularization encourages a neural network to learn like Student B—focusing on patterns that transfer to new situations rather than memorising every detail of the training data

Human-Centred Discussion

Can overfitting happen in humans?

- Yes/NO/Maybe?

Examples:

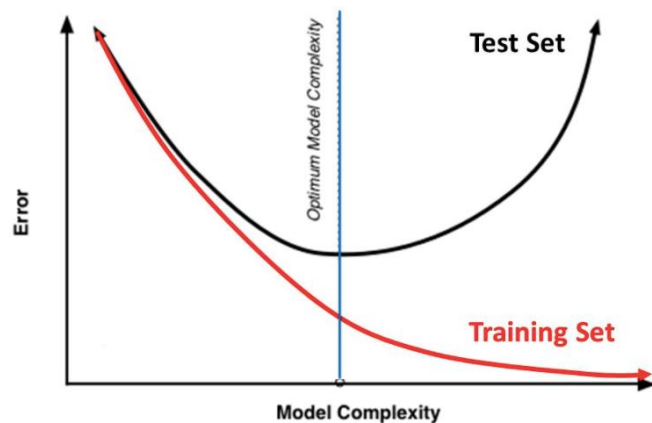
- memorising interview answers
- memorising exam solutions
- memorising driving test routes

GenAI Activity

Ask ChatGPT:

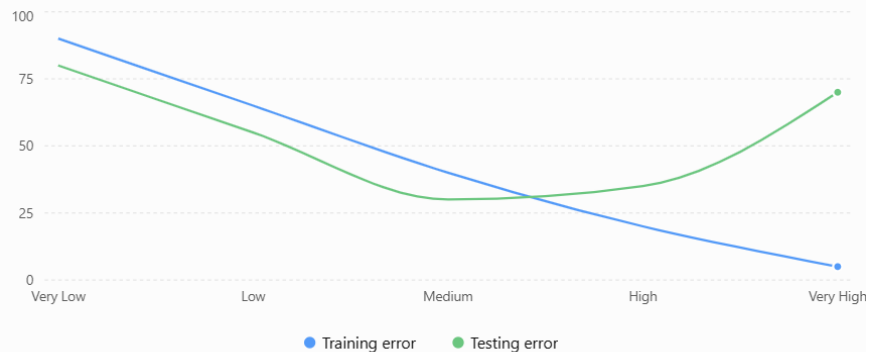
- Explain overfitting using a sports analogy.
- critique
 - clarity
 - correctness
 - missing concepts

Training Vs. Test Set Error



Model complexity vs error

Illustrative relationship between model complexity, training error, and testing error.



What exactly it is?

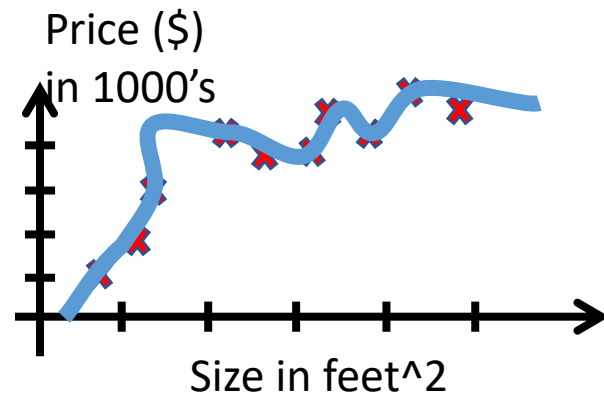
- Imagine interviewing ten people.
- Should you trust someone who memorised every answer,
- or someone who understands the topic?
- Regularization discourages unnecessary complexity.

Human and Deep Learning (Overfitting vs Generalisation)

Human Learning	Deep Learning
Previous exam papers	Training data
Final exam	Test data
Memorising answers	Overfitting
Understanding concepts	Generalisation
Learning principles	Regularization encourages this

Addressing overfitting

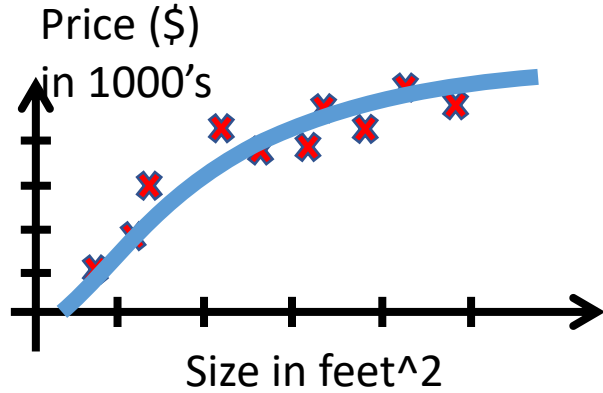
- x_1 = size of house
- x_2 = no. of bedrooms
- x_3 = no. of floors
- x_4 = age of house
- x_5 = average income in neighborhood
- x_6 = kitchen size
- $\vdots$
- x_{100}



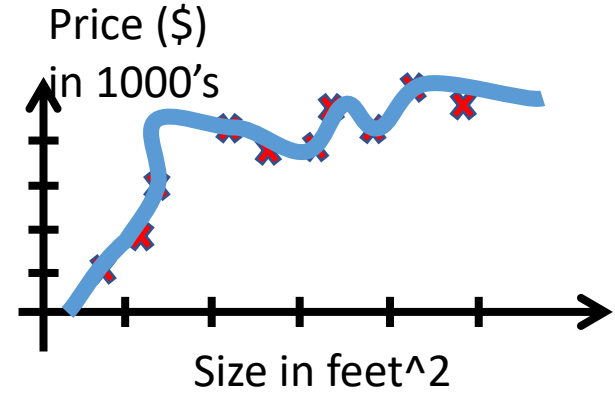
Addressing overfitting

- **1. Reduce number of features.**
 - Manually select which features to keep.
 - Model selection algorithm (later in course).
- **2. Regularization.**
 - Keep all the features, but reduce magnitude/values of parameters θ_j .
 - Works well when we have a lot of features, each of which contributes a bit to predicting y .

Regularization Cost Function – Intuition



$$h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$



$$h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

- Suppose we penalize and make θ_3, θ_4 really small.

$$\min_{\theta} J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + 1000 \theta_3^2 + 1000 \theta_4^2$$

Regularization.

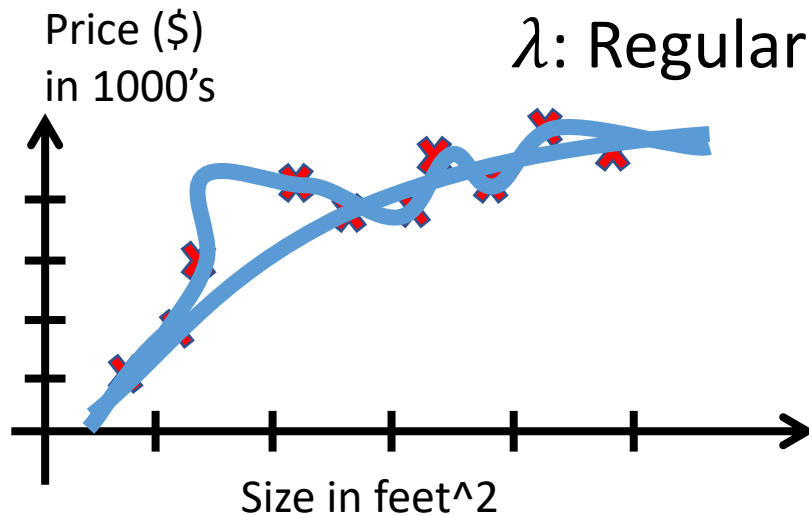
- Small values for parameters $\theta_1, \theta_2, \dots, \theta_n$
 - “Simpler” hypothesis
 - Less prone to overfitting
- Housing:
 - Features: $x_1, x_2, \dots, x_{100}$
 - Parameters: $\theta_0, \theta_1, \theta_2, \dots, \theta_{100}$

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

Regularization

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

$\min_{\theta} J(\theta)$



λ : Regularization parameter

Question

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

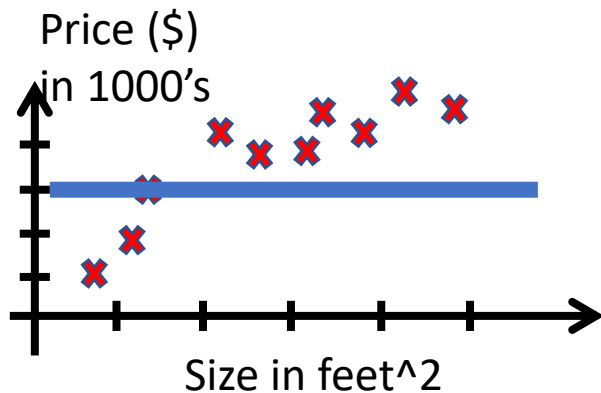
What if λ is set to an extremely large value (say $\lambda = 10^{10}$)?

1. Algorithm works fine; setting to be very large can't hurt it
2. Algorithm fails to eliminate overfitting.
3. Algorithm results in underfitting. (Fails to fit even training data well).
4. Gradient descent will fail to converge.

Question

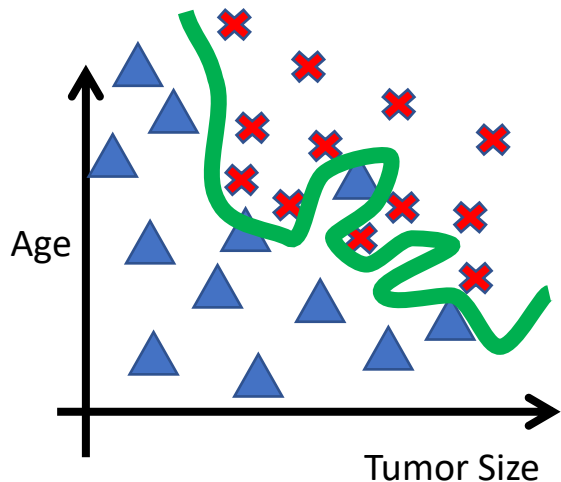
$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

What if λ is set to an extremely large value (say $\lambda = 10^{10}$)?



$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n = \theta^T x$$

Regularized logistic regression



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2 + \theta_6 x_1^3 x_2 + \theta_7 x_1 x_2^3 + \dots)$$

- Cost function:

$$J(\theta) = \frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] + \frac{\lambda}{2} \sum_{j=1}^n \theta_j^2$$

Gradient descent (Regularized)

Repeat {

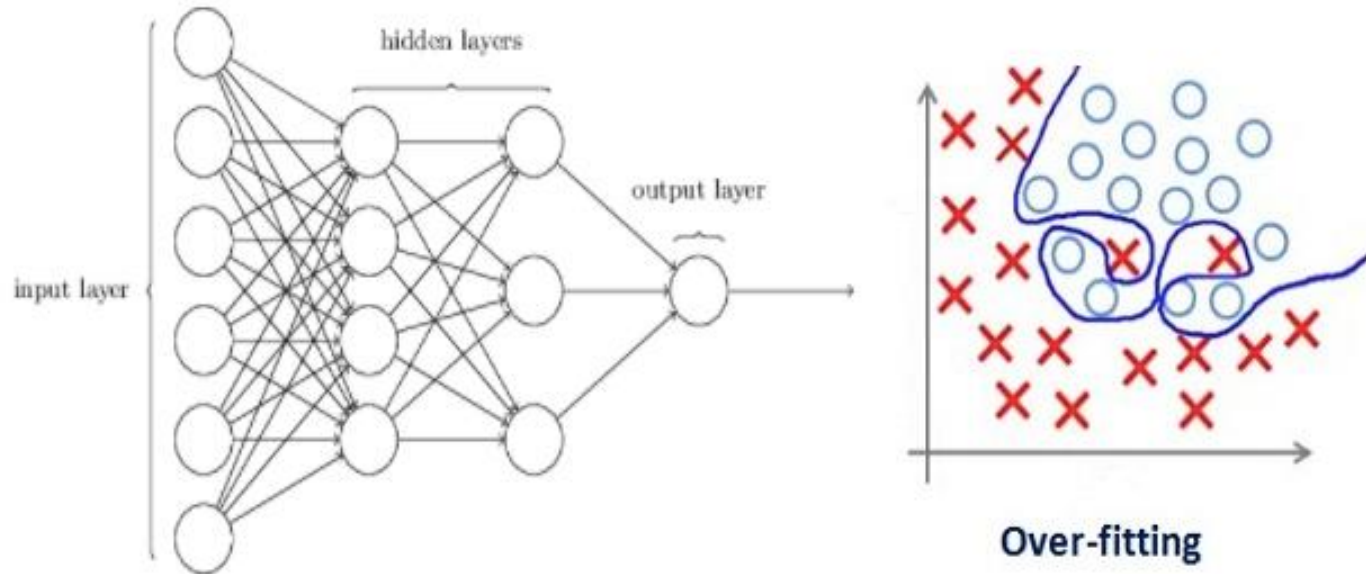
$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$\theta_j := \theta_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} - \lambda \theta_j \right]$$

}

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

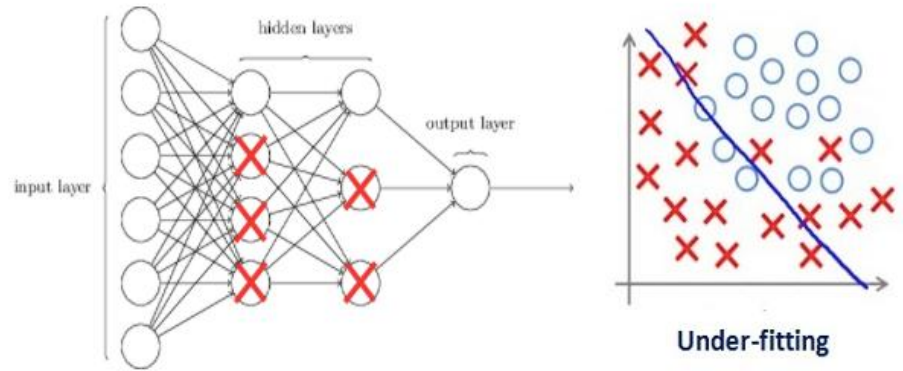
$$\frac{\partial}{\partial \theta_j} J(\theta)$$



Regularized Neural Networks why?

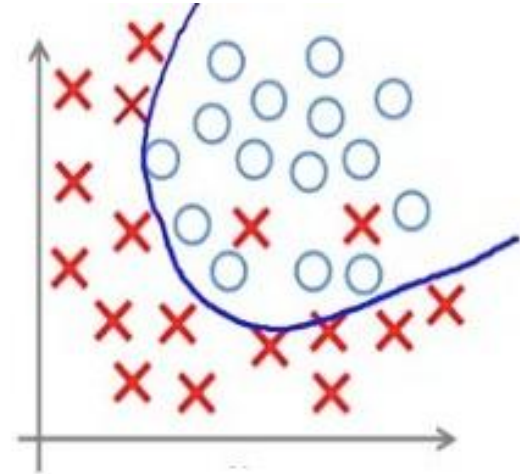
Regularized Neural Networks why? (Cont...)

- If you have studied the concept of regularization in machine learning, you will have a fair idea that regularization penalizes the coefficients. In deep learning, it actually penalizes the weight matrices of the nodes.
- Assume that our regularization coefficient is so high that some of the weight matrices are nearly equal to zero.



Regularized Neural Networks why? (Cont...)

- This will result in a much simpler linear network and slight underfitting of the training data.
- Such a large value of the regularization coefficient is not that useful. We need to optimize the value of regularization coefficient in order to obtain a well-fitted model as shown in the image below.



Appropriate-fitting

L2 and L1 Regularisation



- **Imagine your bedroom is messy. You want to organise it. What would you do?**
- Put things away.
- Throw away unnecessary items.
- Rearrange furniture.
- Clean the room.
- **Therefore, L1 and L2 regularization are like two different ways of cleaning your room.**

L2 Regularization	L1 Regularization
Keep everything	Remove unnecessary features
Shrink weights	Some weights become exactly zero
Smoother model	Simpler model
Uses all features	Performs feature selection

Regularization Terms

- *Cost function = Loss (say, binary cross entropy) + Regularization term*

- In L2
$$\text{Cost function} = \text{Loss} + \frac{\lambda}{2m} * \sum \|w\|^2$$

- In L1
$$\text{Cost function} = \text{Loss} + \frac{\lambda}{2m} * \sum \|w\|$$

L2: *Encourage the student to use **all** their knowledge but not rely too heavily on any single idea. (Balanced understanding.)*

L1: *Help the student identify and **discard irrelevant information**, focusing only on what truly matters. (Selective understanding.)*

$$J(\Theta) = -\frac{1}{m} \left[\sum_{i=1}^m \sum_{k=1}^K y_k^{(i)} \log(h_{\Theta}(x^{(i)}))_k + (1 - y_k^{(i)}) \log(1 - (h_{\Theta}(x^{(i)}))_k) \right] + \frac{\lambda}{2m} \sum_{l=1}^{L-1} \sum_{i=1}^{s_l} \sum_{j=1}^{s_{l+1}} (\Theta_{ji}^{(l)})^2$$

Regularization in Forward and Backpropagation

An Overview of Regularization Techniques in Deep Learning (with Python code)

- This is a great blog post for deep learning beginners, as it provides a summarised presentation of the regularization concept. This blog highlights all the basic points that you need to know to understand regularization.
- **Activity** – This post also contains a case study. Pay extra attention to the case study, as it will be used in one of the in-class activities. It is highly recommended that you download all the necessary software and data related to this case study. ***You must reproduce this work in Python for your understanding.***
- **Reference:** Shubham, J. (2018, 19 April). *An overview of regularization techniques in deep learning (with Python code)* [Web log post]. Retrieved from <https://www.analyticsvidhya.com/blog/2018/04/fundamentals-deep-learning-regularization-techniques/>

Key Takeaways

- A good model should generalise, not memorise.
- Overfitting performs well on training data but poorly on unseen data.
- Underfitting occurs when a model is too simple to learn meaningful patterns.
- Regularization reduces overfitting by controlling model complexity.
- L2 shrinks all weights; L1 removes less important features.
- The best AI models balance accuracy, generalisation, and human judgement.

Learning Activity: Discussion Forum Post

- Using the deep learning resources, you have read in this Module, please identify at least two projects or research ideas that you would like to pursue as part of this unit.
- Describe each of these in one clear problem statement and write associated challenges or research questions for each statement. In this activity, you should demonstrate your ability to identify a research gap, propose a project to address the gap and communicate details in short written form. It should also help you to connect with likeminded people in the class.
- **Post your ideas to the Module 1 discussion forum.**
- **Please read other students' posts and provide feedback on their ideas. Most importantly, identify other students who are planning to work on projects similar to yours.**